INSTITUTE OF SONOLOGY

Royal Conservatoire The Hague

Effects

Programming and Music

http://www.bjarni-gunnarsson.net

Sampling

Music derived from ***recorded sounds*** and the handling of these sounds.

The discussion will focus on:

* *Recording sounds*
* *Storing recorded sound*
* *Playing back recorded sounds*
* *Treatments of recorded sounds*

Playing back

Sounds are usually loaded to buffers and played from there. Another approach is to stream sounds directly from disk. The first approach offers more flexibility but the second one consumes less memory resources.

Important parameters when playing back sounds are:

- ***Speed*** of playback
- ***Direction*** the samples is read in
- ***Location*** where the reading takes place

SuperCollider allows for various ways to read buffer contents at different speeds, directions and locations.

Processing

Alteration of sound signals through designed processes.

The following will be discussed:

* *Delay*
* *Reverb*
* *Resonance*
* *Distortion*
* *Waveshaping*
* *Dynamics*
* *Phase vocoding*
* *Resonators*

Delay

Delay effects can be categorized to ***fixed*** and ***variable*** delay effects. For fixed the delay time does not change. Three different thresholds can be defined for fixed delay times:

* ***Short*** (less than 10 ms) mostly used for frequency-domain purposes for example to create filter.

* ***Medium*** (10 to 50 ms) can give power to a 'thin' signal and create resonances.

* ***Long*** (greater than 50 ms) produce echo outputs, repetitions and bouncing effects.

Delay

Variable time delay effects take place when the delay time is constantly changing.

Examples for variable delays are from effects of the 1960s and 1970s:

* ***Flanging*** is when noise is heard in a mixture of direct and delayed sound. It is created by varying the delay time between 10 to 30 ms with a sine wave 0.1 to 20hz.

* ***Phasing*** is similar to flanging but introduces filters to the signal chain and therefore a slightly less harsh sound.

* ***Chorusing*** consists of bank of variable delays with a slightly longer delay time than the ones mentioned above.

Reverb

Used to add ***impression of space*** to a sound source.

Reverb can be divided to three phases:

* ***Direct sound***
* ***Early reflections***
* ***Gradual reverberation***

The most common reverb technique in digital acoustics is to use ***parallel comb delays*** that are fed into few ***allpass delays***.

Reverb

Popular parameters for controlling reverb algorithms are often:

* Reverb ***decay time.***

* ***Size*** of emulated space.

* High- and Low-frequency ***damping*** parameters.

* ***Pre-delay*** time.

* Amount of ***early reflections.***

Convolution

Convolution is used both for ***filters*** and ***impulse response based reverbs***.

The convolution algorithm ***multiplies*** an incoming source with an impulse response and transfers the characteristics of the impulse response onto the source sound.

Two main methods exist to record impulse responses, by recording an ***impulse-like sound*** (clapping or exploding a ballon) in the corresponding space or ***playing back a sine-sweep*** covering the whole frequency spectrum.

Other applications include ***transferring the spectral envelope*** (or other attributes) from one sound to the other with ***cross-synthesis***.

Distortion

Distortion (or warping) is the alteration of an original signal by reducing or shaping it to produce harshness or overtones.

Examples include:

* ***Saturation***, where levels are pushed above the maximum possible range of a waveform which introduces clipping in the resulting sound.

* ***Bit reduction***, where the precision of a signal is lowered to such an extend that noises appear.

* ***Feedback distortion***, where sound is fed back into the signal chain until distortion effects and sidebands result.

Waveshaping

Waveshaping is a type of distortion technique where complex spectra are produced altering the shape of the waveforms with other ***(shaping or transfer) functions***.

The output of a waveshaping procedure is often a more ***complex*** sound then before and by modulating the ***waveshaping index*** (input amplitude) one can control the generated richness.

The transfer function is often ***nonlinear*** and the output hard to predict.

Compressor

A compressor ***reduces*** the ***dynamic range*** of a signal when it exceeds a certain ***threshold*** of amplitude.

The compression ***ratio*** determines how much the signal that exceeds the threshold is turned down. 2:1 compression ratio means that for every 2 dB of signal that goes into the unit, only 1 dB comes out.

A ***sustainer*** works like an inverted compressor, it exaggerates the low amplitudes and tries to raise them up to the threshold defined.

Limiter

The ***limiter*** operates in a similar way as a compressor does, but looks at the signal's peaks whereas the compressor looks at the average energy level.

A limiter will ***not let the signal past*** the threshold, while the compressor does, using the compression ratio.

Limiters can be considered as more extreme cases of compression with the main difference being the ***lower ratios*** used in compression and the operation of a ***maximum setting*** for a limiter.

Noise Gate

The ***noise gate*** allows a signal to pass through the filter only when it is above a certain threshold.

When the signal level is ***below the threshold***, no sound is allowed to pass.

A noise gate usually has ***attack*** and ***release*** settings to control how the gate kicks in.

Noise gate is often used to ***remove background noise*** but can also create interesting ***rhythmic effects*** as well as to ***cross-modulate*** signals.

Phase Vocoding

Phase vocoding is a form of transformation via analysis where a sound file is analyzed to analysis bands and is then resynthesized by the phase vocoder.

The algorithm allows ***frequency-domain modifications*** to the resulting analysis data (typically time expansion/compression and pitch shifting).

Many possible operations such as:
* ***Shuffling*** of analysis data
* ***Morphing*** two sets of analysis data
* Make ***spectral gaps*** in analysis data
* ***Freeze*** spectral segments
* ***Smooth*** out analysis data

Resonator

A resonator ***adds pitch or oscillation*** to an input signal for specified resonant frequencies.

Resonators are usually created ***using short delay lines*** with ***feedback*** and often occur in ***banks*** where several ones are used simultaneously.

Can be used to ***give pitch*** to input sounds, make ***chords*** from input sounds or ***tune sounds*** to the frequencies of others with subtle resonating behaviors.

Exercises

Exercises - Processing

1. Create a reverb based on a network of all pass filters where each component is slightly distorted.

2. Create an echo effect that uses different processing for low and high frequencies.

3. Create an effect based on delays where the delay time is changing.

4. Experiment with a network that includes a resonator, filters and distortion to emphasise important frequencies in an input sound.

Exercises - Sampling

1. Implement a small loop of sampled sound that changes slightly as time progresses.

2. Create a sound process that oscillates between a pitched down sound sample and pitched up one where the oscillation is controlled by a slow sine wave.

3. Implement a pattern that will play chords using sound samples as base material.

4. Implement a pattern that will play bits of a sound file somehow randomly while starting at the beginning and progressing towards the end of the file.